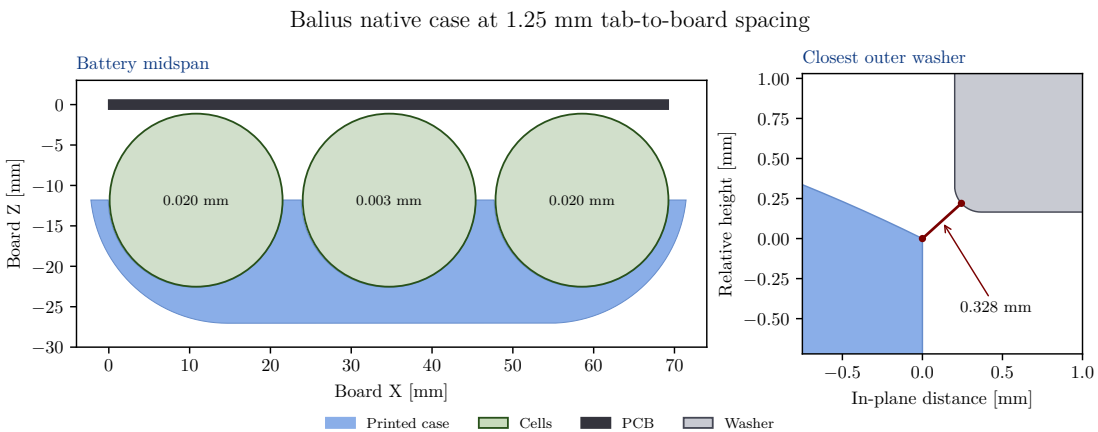
The three-pocket Balius case fits at 1.25 mm stand-off



The arch around the middle cell comes within 0.003 mm. The closest outer washer clears by 0.328 mm. These are rigid distances in the registered SolidWorks geometry.

The shell adds 73.7 g to the 251.8 g bare assembly

Snug geometry

Three cells, four 6.35 mm thick tabs and four M4 case screws. The nominal model uses printed PLA, an air gap, and either nylon or aluminium screws.

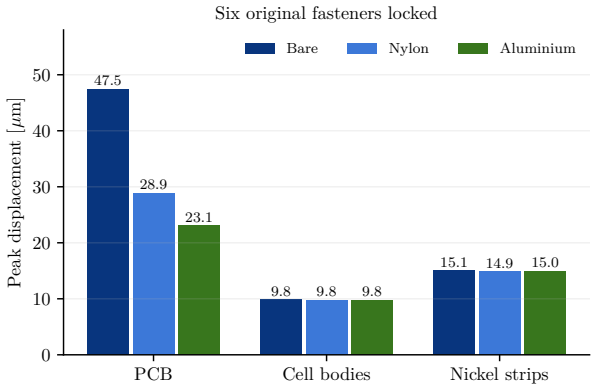
The shell mass assumes dense PLA. Screws and spacer mass are excluded from the estimate.

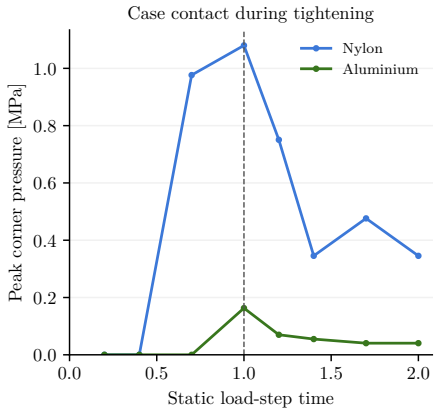
No-case reference

First prestressed mode 485.96 Hz. Peak board-normal motion under base Z was 52.67 μm RMS. At 0.50 mm lateral travel the bare fixture required 8.54 N in positive X and 8.51 N in negative X.

The rerun compares the same loads and movie subjects. The earlier flush and 3 mm results are preserved in Appendix A of the report.

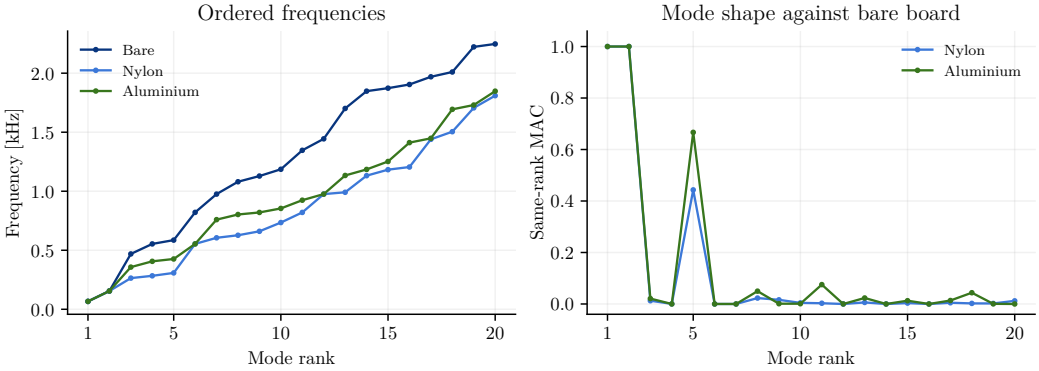
Both snug mounts reduce PCB movement during tightening





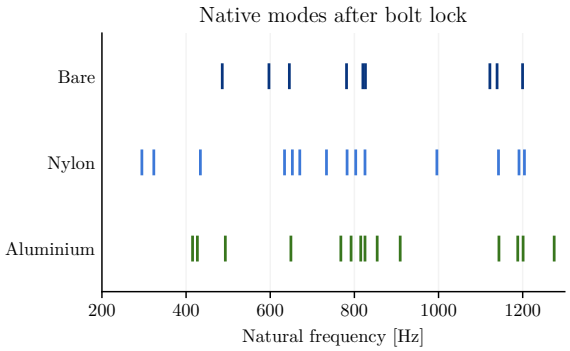
At bolt lock, peak PCB displacement fell from **47.46 μm bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. Only the middle cell contacted the case. A second nylon mesh reproduced PCB deflection within 0.016%. Its peak pocket pressure at lock differed by a factor of two.

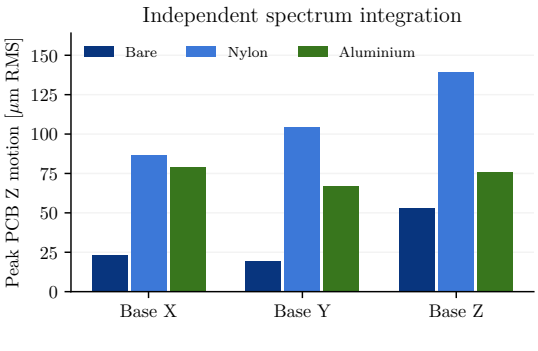
The first unpreloaded mode stays at 66.61 Hz



The first two mode shapes closely match the bare board. At mode 20 the shapes differ, so comparing frequencies by rank alone would give a misleading shift. These are the refined default-contact controls before bolt tightening. Paired movies.

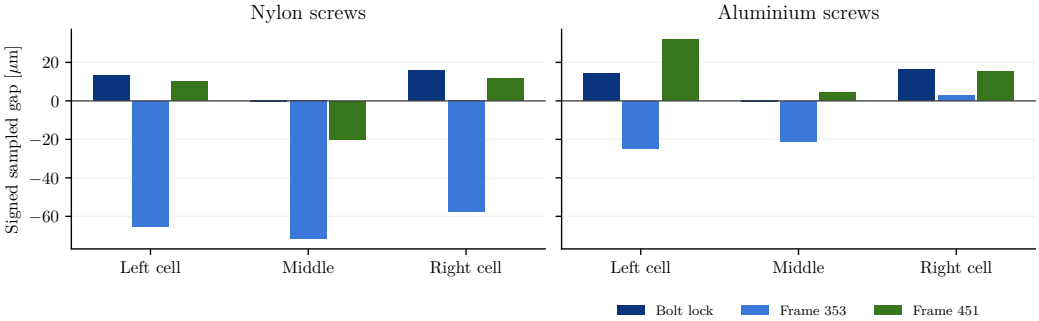
Both snug mounts increase board motion under vibration





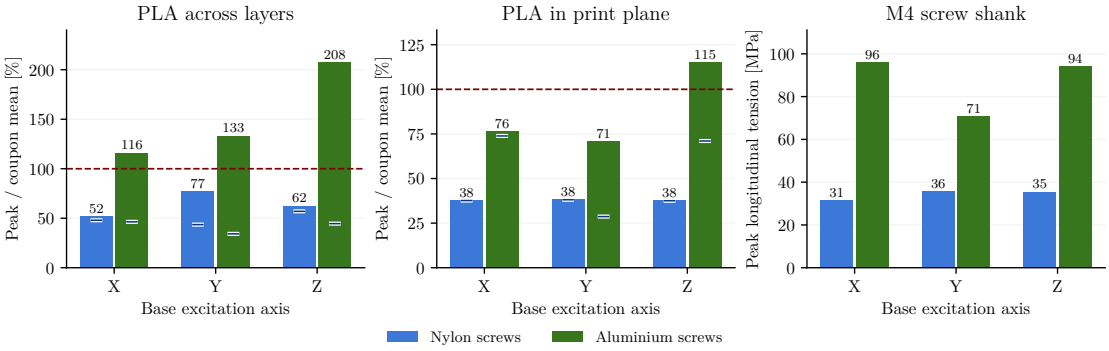
Under base Z, peak board-normal motion rises from **52.7 μm RMS bare** to **139.5 with nylon** and **75.9 with aluminium**. The spectrum calculation holds the installed contact state fixed. The two cased first modes have similar shapes at 295 and 415 Hz.

The outer cells can reach the case



At 1.02155 s under base Z, the linear reconstruction puts all three cells into the nylon-mounted case and two into the aluminium-mounted case. At 1.02753 s, all three clear the aluminium-mounted case. Contact forces require a nonlinear check.

The printed tab exceeds its coupon reference with aluminium screws



The tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**, both 0.60 mm from a tied bearing node. Horizontal marks show the maxima outside the 2 mm bearing band. All three cells reached the case at both peak instants in the linear reconstruction.